

System Design Basics

Lesson 5: Putting It All Together: Design a Simple System

Putting It All Together: Design a Simple System - Study Notes

Key Concepts

- Requirements gathering
- Component diagram
- Data flow and API contracts
- Scalability and reliability strategies

Summary

In this capstone lesson, you synthesize previous lessons to design a photo sharing application. Start by defining functional requirements (upload, view, like) and non functional ones (scalability, availability). Draft a component diagram: API gateway, authentication service, image storage, metadata database, and CDN. Choose PostgreSQL for metadata, S3 for images, and Redis for caching. Apply load balancing, retries, and circuit breakers to ensure reliability. Finally, outline monitoring and scaling rules.

Important Points

- Separate concerns into distinct services.
- Use a CDN to serve static content.
- Implement health checks for automated recovery.

Example

API gateway routes /photos to ImageService; ImageService stores file in S3, writes metadata to Postgres, and updates Redis cache.

Quick Review

- What services are needed for a photo sharing app?
- How would you scale the image storage layer?

